Viscosity of a Non-Newtonian Fluid

Pre-Lab Plan
Unit Operations Lab 5
10/29/2025

Group 3, Tuesday AM section

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1. Experimental Objective (5 pts)

This experiment aims to explore the rheological behavior of a 1 wt% non-Newtonian Carbopol solution. This is done by gauging its flow attributes through various capillary tubes and applying the power law model, otherwise known as the Ostwald-deWaele model. This is all done to compute the flow consistency index, m, and flow behavior index, n, both of which characterize the fluid's viscosity as a function of shear rate. By looking at varying shear conditions, the deviations between Newtonian and non-Newtonian fluids are presented.

2. Experimental

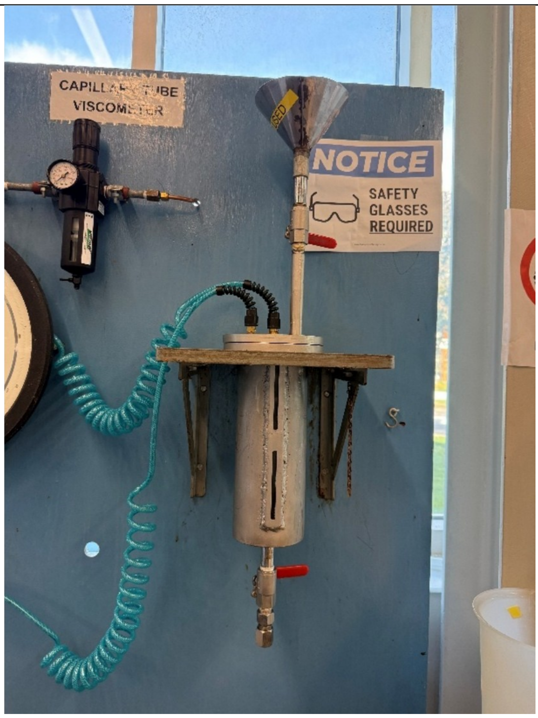
2.1 Apparatus (5 pts)

The apparatus consists of a sealed pressurization vessel with a small capillary tube attached to the bottom. The capillary tubes are removable and can be changed to different sizes. Air then enters through the pressure regulator, pushing the non-Newtonian Carbopol solution through the capillary. The pressure gauge then reads the pressure inside the pressurization vessel as the fluid flows through the capillary tube.

Table 2.1: List of Main Apparatus Equipment and Descriptions

Item	Picture	Description
Pressure gauge valve		Allows for the isolation of pressure out of the regulator to the rest of the experiment

Pressure gauge		Measures the pressure in the viscometer, max pressure is 200 psig.
Pressure regulator		Regulates pressure applied to the vessel, max inlet and outlet pressure is 150 psig.

Viscometer vessel		Vessel that holds the Carbopol solution
Vent valve		Depressurizes the vessel

valve		Used to start and stop fluid flow out of the vessel
Capillary tubes		Tubes in varying lengths and diameters that are attached to the bottom of the vessel

2.2 Materials and Supplies (5 pts)

Table 2.2: List of Materials and Supplies

Material/ Supply	**Description**
Carbopol 934 Powder	Polymer used to create non-Newtonian fluid
0.1 wt% NaOH	Used to neutralize the Carbopol solution
pH paper	Used to ensure Carbopol solution has a pH between 5-7
4L beaker	Vessel to make our solution in
2L bucket	Vessel to store solution
Graduated	Used to measure the density of the solution

cylinders	
Allen wrench	Used to screw bolts in to seal the vessel
Scale	Used to measure Carbopol powder
Electric mixer	Used to mix Carbopol solution
Spatula	Used to scrape down sides when mixing the solution
Gloves	PPE as Carbopol is slightly acidic
Paper towels	Used to clean apparatus
Silicone sealant	Used to ensure a good seal on the vessel
Compressed air	Used to pressurize the vessel
Lab jack	Used when mixing the solution
DI water	Used to make solution

2.3 Experimental Plan (10 pts)

Procedure:

- Prepare a 2L solution of 1 wt% Carbopol in DI water and mix until fully dissolved
- Gradually neutralize the solution using NaOH until a pH of 5-7 is reached
- Mount the capillary, and fill the reservoir with the prepared solution
- Set ΔP to 3-5 levels from 0-60 psig
- Measure Q, T, time, and collected mass/volume
- Take 1 set of replicate data for error analysis

Variables:

- Independent: Pressure Drop (0-60 psig) and capillary type
- Dependent: Q, shear stress, shear rate, and viscosity

Statistical and Calibration Methods:

- Using a known standard, verify the pressure gauge
- Take replicate data
- Least-squares regression can be used for goodness of fit
- Average replicate Q values and calculate SD and 95% confidence intervals

Theory Correlation:

- This experiment applies the Ostwald-deWaele power law model

$$\tau_{yx} = \pm \left| \frac{d v_x}{dy} \right|^n$$

- Plot shear stress vs shear rate to classify the fluid

Project Safety Evaluation

Project Safety Evaluation (10 pts)

	Yes/No	
Potential electrical hazards?	yes	If yes, identify conditions: The mixer and its controller are prone to water damage. Another hazard is getting caught in the mixer itself.
Potential mechanical hazards?	yes	If yes, identify conditions: Capillary tubes can be poorly secured to the viscometer, under which pressure applied can cause them to come apart.
Potential pressure hazards?	yes	If yes, identify conditions: The entire system should be lower than 90psig from the house air supply after the regulator. The facility air pressure coming in should also be coming in at around 60psig.
Potential temperature hazards?	no	If yes, identify conditions:
Hazardous/toxic chemicals involved?	yes	If yes, hazards are involved: Sodium hydroxide is a corrosive, posing a risk to skin contact. Carbopol also poses a risk, as it can cause respiratory irritation.
Gloves required?	yes	If yes, specify type: Gloves should be used whenever handling Sodium hydroxide and Carbopol.
MSDS reviewed by all team members?	yes	List MSDS sheets reviewed: Water(7732-18-5), Carbopol 934(9003-01-4), 0.1% NaOH (9003-01-4)
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	25	
Pressure (psia)	104.7	
Safety Controls	*yes/no*	*If yes, Provide description*
	yes	*Pressure regulators allow us to control the pressure coming out and into the system.*

I have read relevant background material for the Unit Operations Laboratory entitled:
“____” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Aaron Casas		10-29-25
	signature of team member	date
Anna Cai		10-29-25
	signature of team member	date
Hussein Rezk		10-29-25
	signature of team member	date
Adil Feratovic		10-29-25
	signature of team member	date